

MUSIC STORE DATA ANALYSIS

SQL Project Report

MySQL • Relational Data Analysis • Business Insights

Scope: 8 Business level question sets

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1. Executive Summary

This report consolidates the eight Music Store SQL questions completed so far using music_db in MySQL Workbench. It records the SQL logic, the results shared during the project, and practical interpretations.

The analysis covers employee hierarchy, invoice distribution by country, highest invoice values, city-level revenue, customer spending, Rock-music purchasers, artists with the most Rock tracks, and tracks longer than average.

2. Project Objectives

- Answer practical business questions using SQL on a relational database.
- Practice SELECT, filtering, sorting, aggregation, GROUP BY, JOIN, DISTINCT, LIMIT, and subqueries.
- Translate query outputs into cautious, evidence-based business observations.
- Create reusable documentation for a portfolio or GitHub project.

3. Database Overview

The music_db schema discussed in the project contains 11 tables. The diagram summarizes the relationships discussed while building the model. It is a recreated EER-style overview, not a direct screenshot exported from MySQL Workbench.

Table	Purpose	Key relationships
employee	Employee and reporting hierarchy	Self-reference reports_to; supports customer
customer	Customer details	Linked to employee and invoice
invoice	Invoice headers and totals	Linked to customer
invoice_line	Invoice line items	Linked to invoice and track
track	Track catalogue	Linked to album, genre, media_type
album	Album catalogue	Linked to artist
artist	Artist catalogue	Parent of album
genre	Genre lookup	Linked to track
media_type	Media format lookup	Linked to track
playlist	Playlist catalogue	Linked through playlist_track
playlist_track	Playlist-track bridge	Links playlist and track

3.1 EER Relationship Diagram

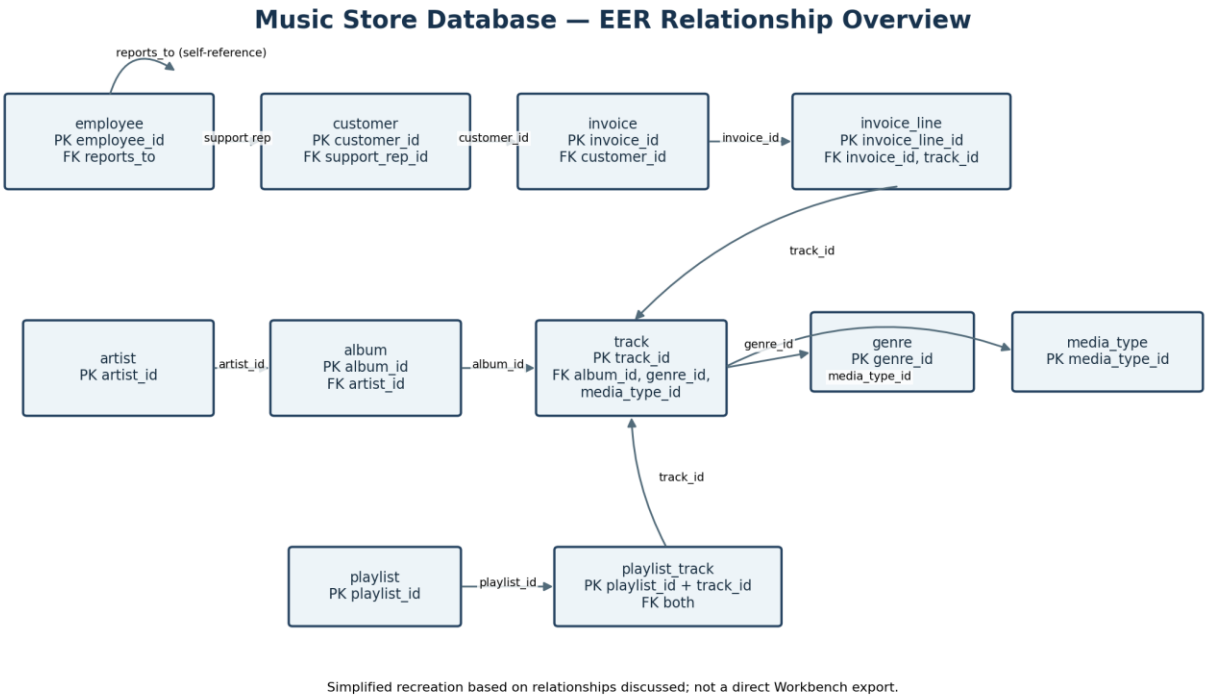


Figure 1. Simplified EER relationship overview.

3.2 Main Relationships

- employee.reports_to references employee.employee_id.
- customer.support_rep_id references employee.employee_id.
- invoice.customer_id references customer.customer_id.
- invoice_line.invoice_id references invoice.invoice_id; invoice_line.track_id references track.track_id.
- album.artist_id references artist.artist_id.
- track references album, genre, and media_type.
- playlist_track connects playlist and track.

4. Analysis and Results

4.1 Question 1 — Senior-most employee

Business question: Who is the senior-most employee based on job title?

SQL query

```
SELECT first_name, last_name, levels
FROM employee
ORDER BY levels DESC;
```

Reported hierarchy: Mohan Madan — L5; Andrew Adams — L4; Nancy Edwards and Jane Peacock — L3; Margaret Park and Steve Johnson — L2; Michael Mitchell, Robert King, and Laura Callahan — L1.

Mohan Madan has the highest level shown: L5.

Business interpretation

- The levels column can be sorted to inspect the recorded hierarchy.
- Mohan Madan is the highest-ranked employee according to the supplied levels.

Note: The question mentions job title, but the submitted query sorts levels rather than title. This finding is based on levels.

4.2 Question 2 — Countries with the most invoices

Business question: Which countries have the most invoices?

SQL query

```
SELECT billing_country AS Country, COUNT(*) AS total_invoices
FROM invoice
WHERE billing_country IS NOT NULL
GROUP BY billing_country
ORDER BY total_invoices DESC
LIMIT 5;
```

Top five reported: USA — 131; Canada — 76; Brazil — 61; France — 50; Germany — 41. Combined total: 359 invoices.

Business interpretation

- The USA leads the shared invoice counts.
- Invoice count measures transaction frequency, not necessarily revenue.

4.3 Question 3 — Top three invoice totals

Business question: What are the top 3 values of total invoice?

SQL query

```
SELECT *
FROM invoice
ORDER BY total DESC
LIMIT 3;
```

Reported results: Invoice 183, customer 42, France — 23.76; invoice 92, customer 32, Canada — 19.80; invoice 526, customer 5, Czech Republic — 19.80.

Business interpretation

- The highest invoice value shown is 23.76.
- Invoices 92 and 526 tie at 19.80.

Note: This returns three invoice records, not the three highest distinct values. Equal totals may appear in any order without a secondary sort key.

4.4 Question 4 — City generating the most revenue

Business question: Which city generates the most revenue?

SQL query

```
SELECT billing_city, ROUND(SUM(total), 2) AS total_revenue
FROM invoice
GROUP BY billing_city
ORDER BY total_revenue DESC
LIMIT 1;
```

Reported result: Prague — 273.24 total invoice revenue.

Business interpretation

- Prague has the highest sum of invoice totals in the supplied result.
- Revenue alone does not establish profitability or prove festival suitability.

Note: Currency was not confirmed in the shared output, so the number is preserved as displayed.

4.5 Question 5 — Best customer by spending

Business question: Who has spent the most money?

SQL query

```
SELECT c.customer_id, c.first_name, c.last_name,
       ROUND(SUM(i.total), 2) AS total_revenue
FROM customer AS c
JOIN invoice AS i ON c.customer_id = i.customer_id
GROUP BY c.customer_id
ORDER BY total_revenue DESC
LIMIT 1;
```

Reported result: Customer 5 — František Wichterlová — 144.54 total spending.

Business interpretation

- JOIN connects customers to invoices; SUM calculates spending across invoices.
- The highest-spending customer in the shared result is František Wichterlová.

Note: The original output displayed character-encoding artifacts. The name is rendered here with intended accents; check database encoding if this persists.

4.6 Question 6 — Customers who purchased Rock music

Business question: Return email, first name, last name, and genre for Rock-music listeners, ordered by email.

SQL query

```
SELECT DISTINCT c.email, c.first_name, c.last_name, g.name AS genre
FROM customer AS c
JOIN invoice AS i ON c.customer_id = i.customer_id
JOIN invoice_line AS il ON i.invoice_id = il.invoice_id
JOIN track AS t ON il.track_id = t.track_id
JOIN genre AS g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
ORDER BY c.email ASC;
```

Sample rows shared: aaronmitchell@yahoo.ca — Aaron Mitchell; alero@uol.com.br — Alexandre Rocha; astrid.gruber@apple.at — Astrid Gruber; bjorn.hansen@yahoo.no — Bjørn Hansen; camille.bernard@yahoo.fr — Camille Bernard. Complete result count was not provided.

Business interpretation

- Joins trace purchases to tracks and genres.
- DISTINCT avoids repeated customer/genre rows when multiple Rock tracks were purchased.
- The query can support analysis of customers who purchased Rock tracks.

Note: The database records purchases, not actual listening behavior. Email domains do not reliably prove customer location.

4.7 Question 7 — Artists with the most Rock tracks

Business question: Return the top 10 artists by number of Rock tracks.

SQL query

```
SELECT a.name AS artist_name, COUNT(t.track_id) AS total_rock_tracks
FROM artist AS a
JOIN album AS al ON a.artist_id = al.artist_id
JOIN track AS t ON al.album_id = t.album_id
JOIN genre AS g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY a.artist_id, a.name
ORDER BY total_rock_tracks DESC, artist_name ASC
LIMIT 10;
```

Rows shared: AC/DC — 18; Aerosmith — 15; Audioslave — 14; Led Zeppelin — 14; Alanis Morissette — 13; Alice In Chains — 12; Frank Zappa & Captain Beefheart — 9; Accept — 4. Eight rows were supplied.

Business interpretation

- AC/DC leads the shared artist rows with 18 Rock tracks.
- Audioslave and Led Zeppelin tie at 14.
- Track count reflects catalogue size, not necessarily sales or revenue.

Note: If MySQL returned only eight rows, check the full result grid and whether other artists have Rock tracks; the shared output may be partial.

4.8 Question 8 — Tracks longer than average

Business question: Find tracks longer than the average track duration, longest first.

SQL query

```
SELECT name, milliseconds
FROM track
WHERE milliseconds > (SELECT AVG(milliseconds) FROM track)
ORDER BY milliseconds DESC;
```

Reported average: 251,177.7431 milliseconds, approximately 4 minutes 11 seconds.

Rows shared: How Many More Times — 711,836 ms; Advance Romance — 677,694 ms; Sleeping Village — 644,571 ms; You Shook Me(2) — 619,467 ms; Talkin' 'Bout Women Obviously — 589,531 ms; Stratus — 582,086 ms; No More Tears — 555,075 ms; The Alchemist — 509,413 ms; Wheels Of Confusion / The Straightener — 494,524 ms; Book Of Thel — 494,393 ms.

Business interpretation

- The subquery calculates the average; the outer query filters tracks above it.
- How Many More Times is the longest track in the shared rows, at about 11 minutes 52 seconds.
- Duration alone does not establish popularity or commercial success.

Note: The shared table contains nine visible rows despite being described as ten. The query has no LIMIT and returns all tracks above average.

5. Consolidated Findings

Area	Finding
Employee hierarchy	Mohan Madan is L5, the highest level shown.
Invoice distribution	USA leads with 131 invoices, followed by Canada (76) and Brazil (61).
Largest individual invoice	Invoice 183 is 23.76 in the shared top-three results.
City revenue	Prague has the highest summed invoice total in the output: 273.24.
Customer spending	Customer 5, František Wichterlová, has reported spending of 144.54.
Rock purchases	Query identifies customers who purchased Rock tracks, sorted by email.
Rock catalogue	AC/DC leads the shared artist rows with 18 Rock tracks.
Track duration	Average is about 4 minutes 11 seconds; longest shared row is about 11 minutes 52 seconds.

6. Limitations and Data Quality Notes

- Report uses outputs shared during the project; it does not independently query the live database.
- Some outputs were partial: five sample Rock-customer rows, eight artist rows, and nine visible long-track rows.
- The employee query sorted levels rather than title.
- Currency was not confirmed in the supplied invoice outputs.
- Character encoding may affect accented names.
- Invoice count, invoice value, track count, and track duration are different measures and are not direct measures of profit or demand.
- The EER diagram is a recreated overview from known relationships, not a direct Workbench export.

7. Conclusion

The completed Easy and Moderate questions demonstrate practical SQL skills: joins, filtering, sorting, grouping, aggregation, duplicate removal, and subqueries. The results provide an initial view of invoice distribution, city revenue, customer spending, Rock purchases, artist catalogue counts, and track duration.

Appendix A — SQL Concepts Practised

- SELECT and aliases
- WHERE and IS NOT NULL
- ORDER BY ASC/DESC
- LIMIT
- COUNT(), SUM(), AVG(), ROUND()
- GROUP BY
- INNER JOIN
- DISTINCT

